Class 16: Introduction to the Drude-Sommerfeld model

Our intention is to identify theories that match experimental data on material properties. The Drude model was only partially successful in this. In fact, it was partially successful only due to a stroke of luck that certain errors in the model canceled each other in specific instances. After recognizing the Drude models limitations, the search for the cause of its limitations led us to the idea that we had to treat electrons as quantum mechanical particles rather than classical particles. It is in this context that we looked at the history of quantum mechanics, the key concepts of quantum mechanics, and how the key concepts related to each other.

Quantum mechanical effects are relevant in the context of the systems we are studying. At the same time, we must note, that quantum mechanics is not inconsistent with the macroscopic world. It is just that at large size scales quantum mechanical effects become insignificant and hence can be reasonably ignored.

At this time let us introduce the scientist who made the contribution that significantly improved the Drude model – Arnold Sommerfeld. To understand how significant his contributions to science in general have been, let us briefly look at the people shown in Figure 16.1 below and their accomplishments.



Werner Karl Heisenberg

The Nobel Prize in Physics 1932

"for the creation of quantum mechanics, the application of which has, inter alia, led to the discovery of the allotropic forms of hydrogen"

http://nobelprize.org/nobel_prizes/physics/laureates/1932/index.html



Peter Debye

The Nobel Prize in Chemistry 1936

"for his contributions to our knowledge of molecular structure through his investigations on dipole moments and on the diffraction of X-rays and electrons in gases"

http://nobelprize.org/nobel_prizes/chemistry/laureates/1936/

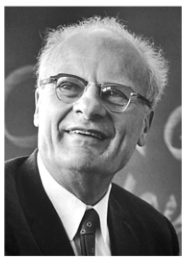


Wolfgang Pauli

The Nobel Prize in Physics 1945

"for the discovery of the Exclusion Principle, also called the Pauli Principle"

http://nobelprize.org/nobel_prizes/physics/laureates/1945/index.html



Hans Bethe

The Nobel Prize in Physics 1967

"for his contributions to the theory of nuclear reactions, especially his discoveries concerning the energy production in stars"

http://nobelprize.org/nobel_prizes/physics/laureates/1967/

Figure 16.1: Four Nobel laureates and their citations

Figure 16.1 above shows four Nobel laureates and the citations for their Nobel Prizes. These awards span a time period of 35 years, and involve work related to subatomic particles, quantum mechanics, and nuclear reactions and energy in stars. Three Nobel Prizes in physics and one in chemistry. What do these four scientists have in common, other than being Nobel laureates? Interestingly, they all had the same PhD guide or advisor – Arnold Sommerfeld. Figure 16.2 below shows a photograph of Arnold Sommerfeld and indicates his contribution which is of interest to us.



Arnold Sommerfeld

Several of his students won Nobel prizes

The Drude-Sommerfeld model

http://www.lrz-muenchen.de/~Sommerfeld/Bilder/as97_01.gif

Figure 16.2: A photograph of Arnold Sommerfeld, and the contribution of his which of immediate interest to us – The Drude-Sommerfeld model.

Arnold Sommerfeld never won the Nobel Prize himself, but made significant contributions to the fields of mathematics and physics. Of immediate interest to us is the Drude-Sommerfeld model, which takes some features of the Drude model and modifies some others, and makes an overall improvement to the Drude model.

The important features of the Drude-Sommerfeld model are listed in Figure 16.3 below.

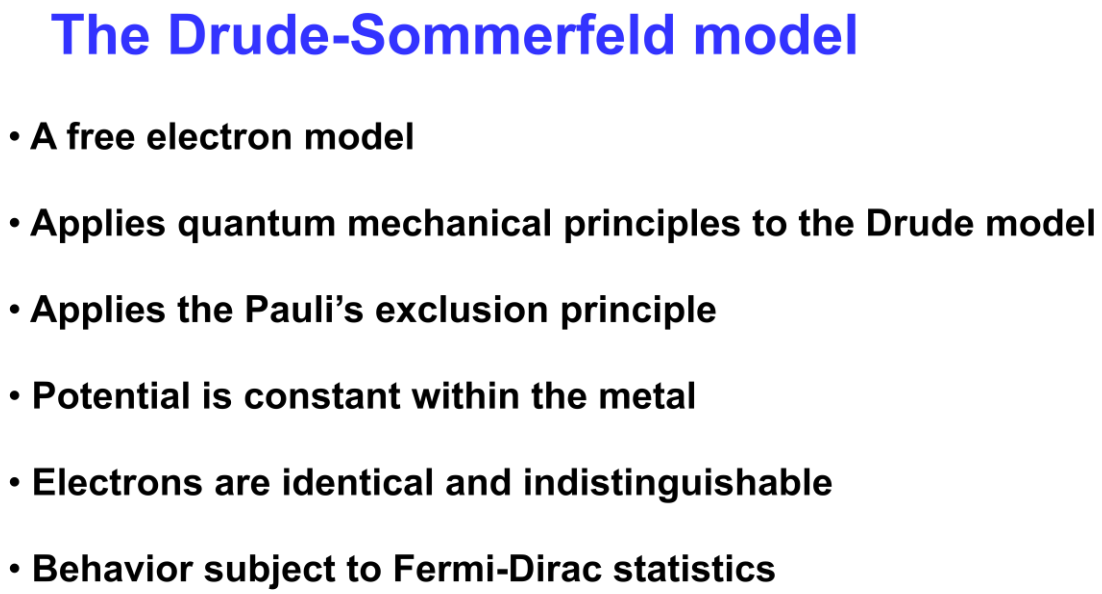


Figure 16.3: Important features of the Drude-Sommerfeld model.

The Drude-Sommerfeld model is a free electron model in the same sense that the Classical Drude model is. This means that the electrons responsible for conduction are not bound to any particular atom, and are free to roam the extent of the solid. The potential within the solid is assumed to be uniform, and the implication of this is that the electrons do not have any preferred site that they are likely to aggregate towards.

In view of the application of quantum mechanical principles, the electrons are assumed to be identical and indistinguishable. This assumption impacts the manner in which the statistics corresponding to the electrons, and their distribution across energy levels, is developed. In addition the assumption made is that the electrons obey the Pauli's exclusion principle.

Identical and indistinguishable particles which obey the Pauli's exclusion principle, follow the statistical description developed by Fermi and Dirac, which is referred to by their names as the Fermi-Dirac statistics. Particles that follow the Fermi-Dirac statistics are called Fermions, in just the manner in which particles following Maxwell-Boltzmann Statistics are called Classical particles. Fermions have the additional characteristic that they possess half integer spins, which the electrons do.

In dealing with Fermions we must recognize that there is the concept of a fixed number of states at any given energy level, which sets an upper limit to the number of electrons that can occupy that energy level. This was not a restriction in the classical Drude model that we discussed earlier. While deriving the Maxwell-Boltzmann statistics we started by saying that let there be n_0 particles at ε_0, n_1 particles at ε_1, n_2 particles at ε_2, n_3 particles at ε_3, n_4 particles at ε_4,...and n_r particles at ε_r. While deriving the Fermi-Dirac statistics we will have to modify that statement and say instead that let there be n_0 particles in s_0 states

at ε_0, n_1 particles in s_1 states at ε_1, n_2 particles in s_2 states at ε_2, n_3 particles in s_3 states at ε_3, n_4 particles in s_4 states at ε_4,...and n_r particles in s_r states at ε_r. If the manner in which we define the states includes all of the quantum numbers, then we can have a maximum of only one particle per state.

In view of the particles being identical and indistinguishable, if the total number of particles at two energy levels remains the same, and a few particles from one energy level are simply swapped for the same number of particles from the other energy level, this does not count as a new microstate for the system.

The limits on the number of particles at a given energy level, and the change in the manner of defining a new microstate for the system, together significantly change the resulting statistics of the system. The Fermi-Dirac statistics therefore provides us with results and predictions that are vastly different from those obtained using the Maxwell-Boltzmann statistics. The important differences between the classical Drude model, and the Drude-Sommerfeld model, are summarized in Figure 16.4 below.

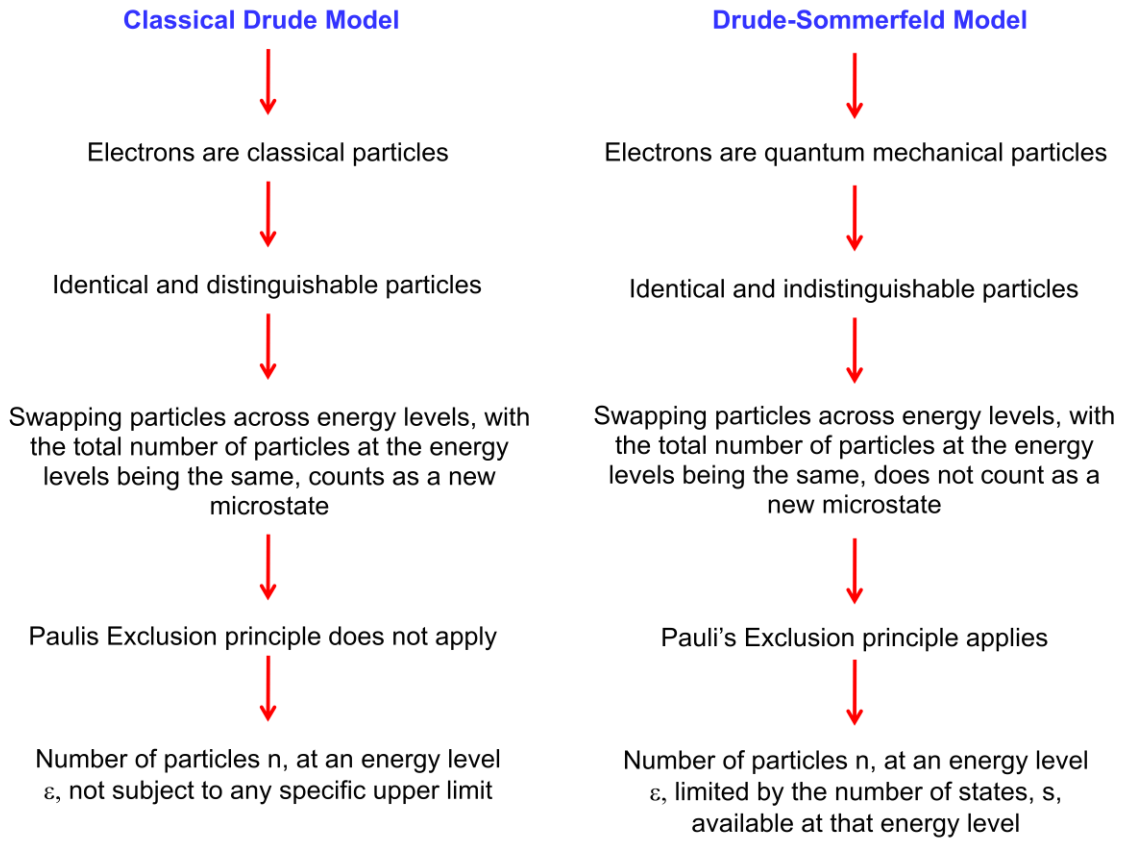


Figure 16.4: Differences between the classical Drude model and the Drude-Sommerfeld model

In the next class we will derive the Fermi-Dirac statistics, and the results of the derivation will tell us if the estimation of the behavior of electrons in a solid has changed.

Specifically we will see if the anomalies in the estimation of C_V^e, and $\langle v^2 \rangle$, have been addressed.